A Study on the Impact of Cost Presentation on Decision-Making Bias and Cancellation Behavior in Online Subscriptions

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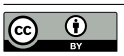
Abstract

Online subscription interfaces often embed cost information, renewal conditions, and cancellation entries within complex page structures, which may increase users' payment misclassification and cancellation abandonment. To strengthen the computational contribution of this study, this paper proposes a cost-presentation bias detection framework that integrates interface structure parsing, visual salience encoding, clickstream sequence modeling, and cancellation path complexity estimation. The proposed framework first extracts DOM hierarchy depth, cost-field screen exposure, price-block coordinates, font scale, contrast ratio, button–cost distance, cancellation entry depth, and hidden-menu expansion from subscription, checkout, and cancellation pages. User behaviors are then synchronized with page-level identifiers and encoded as temporal sequences containing clicks, dwell time, scrolling, back navigation, field expansion, and abandonment events. A structure–behavior fusion model is constructed to estimate session-level decision-bias probability, while a directed path graph is used to compute cancellation-process complexity through node transitions, edge costs, branch entropy, dwell-time accumulation, and semantic jump penalties. Experiments on 4,276 subscription-related pages and 3,892 valid interaction sessions show that deep cost placement increases payment misclassification, whereas upfront cost disclosure, visual salience enhancement, and path compression reduce the mean misjudgment rate from 22.4% to 8.1%. High-complexity cancellation paths lead to a 37.4% abandonment rate, while low-complexity paths reduce the rate to 19.8%. After interface optimization, decision accuracy increases from 77.6% to 91.9%, and unplanned subscription decreases from 12.7% to 6.3%. The results demonstrate that cost-presentation bias can be computationally detected and mitigated through interface-structure modeling and behavior-aware path optimization. This study contributes a measurable HCI-oriented computational method for auditing subscription interfaces and improving digital consumption fairness.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI.

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Keywords

subscription systems, cost disclosure, interface structure, decision-making biases, consumer behavior, digital fairness

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1 Introduction

Online subscription services have become a common payment model in e-commerce, streaming media, cloud software, digital content, and platform-based retail. In these services, users' payment decisions are not determined only by price level, but also by the computationally describable structure through which cost information is displayed. Renewal fees, additional charges, trial-period conversion rules, and cancellation links may appear at different DOM depths, screen positions, visual salience levels, and interaction path locations. When such information is placed outside the first screen, separated from the primary confirmation button, hidden in expandable fields, or embedded in a multi-step cancellation path, users may misclassify the final payment amount or abandon the cancellation process before completion.

Existing studies on subscription behavior mainly explain user retention, switching cost, satisfaction, and consumer response from behavioral or managerial perspectives. However, these studies usually do not provide a machine-readable representation of cost-presentation structure, nor do they model how interface hierarchy, visual salience, and clickstream behavior jointly influence decision bias. As a result, the relationship between cost disclosure design and subscription fairness is often described qualitatively, while the underlying interface mechanism remains difficult to measure, compare, and reproduce. This limitation weakens the applicability of existing findings to interface auditing, automated compliance checking, and human-centered system evaluation.

To address this problem, this paper proposes a computational framework for detecting cost-presentation-induced decision bias and cancellation friction in online subscription interfaces. The proposed framework treats subscription pages not as static visual layouts but as structured computational objects consisting of DOM nodes, visual blocks, action buttons, interaction events, and navigation states. It extracts page-level structural features, including cost-field hierarchy, first-screen exposure ratio, font scale, contrast ratio, price-block location, button–cost distance, cancellation entry depth, hidden expansion count, and back-navigation depth. It

also converts user interaction records into temporal behavior sequences that include click operations, dwell intervals, scrolling, field expansion, backtracking, and abandonment events.

The main contribution of this study lies in three aspects. First, the study constructs a cost-presentation feature extraction pipeline that transforms subscription interface elements into measurable structural and visual variables. Second, the study designs a structure–behavior fusion model that combines interface features and clickstream sequences to estimate session-level decision-bias probability. Third, the study builds a directed cancellation path graph and computes process complexity through node transitions, edge costs, branch entropy, dwell time, and semantic jump penalties. Through experiments on subscription-related interface pages and interaction sessions, this paper evaluates how upfront cost disclosure, visual salience enhancement, and path compression reduce payment misclassification, cancellation abandonment, and unplanned subscription. The study therefore shifts the research focus from general consumer response analysis to a reproducible HCI-oriented computational method for interface bias detection and optimization.

2 Online Subscription Cost Presentation Structure and Behavioral Data Feature Extraction

In the proposed framework, each subscription-related page is processed as a structured computational object rather than as a manually observed webpage. The interface parser first reads the DOM tree and records node type, hierarchy depth, bounding-box coordinates, visibility status, text content, CSS font size, color contrast, button location, hyperlink target, and expandable-panel state. Subscription, checkout, and cancellation pages are then decomposed into six element categories: cost fields, renewal prompts, trial conversion descriptions, primary action buttons, cancellation entries, and secondary confirmation nodes. Cost-related fields are identified through a rule-based lexical matcher that detects price symbols, subscription-cycle terms, renewal keywords, trial-period expressions, cancellation anchors, and recurring-payment phrases. For each detected element, the system calculates the first-screen exposure ratio, DOM depth, coordinate distance from the primary button, normalized font scale, contrast ratio, and hidden-expansion count 1. These indicators are treated as proxy variables of perceived salience rather than direct measurements of visual attention. To reduce the limitation of relying only on heuristic layout features, the framework further incorporates visual-load indicators, including the number of competing clickable elements around the cost block, text density within the first screen, repeated prompt exposure, and the spatial dispersion of cost-related fields. On the behavioral side, dwell-time extension, repeated scrolling, back navigation, and field re-expansion are used as indirect indicators of cognitive load and interface confusion. These structural, visual-load, and behavioral indicators are jointly encoded into the structural feature vector, so that the model does not depend solely on font size or contrast ratio to infer cost visibility.

On the behavioral side, clickstream logs are converted into temporal operation sequences. Each event is represented by event type, timestamp, page identifier, DOM target identifier, scroll offset, dwell interval, back-navigation flag, field-expansion flag, and abandonment label. The sampling granularity is set to 100 ms, and each session captures no more than 300 operation steps. Operation intervals exceeding 2.5 seconds are treated as abnormal pauses and are either clipped or masked during sequence construction. Page-level structural vectors and behavior sequences are aligned by page IDs and timestamps, producing a joint feature matrix for downstream decision-bias recognition and cancellation-path complexity modeling. This processing procedure makes the page layout, visual salience, and user behavior records directly computable and reproducible [2].

3 Design of a Subscription Behavior Analysis Model for Decision Bias Identification

3.1 Framework for Integrating Cost Presentation Structures with User Behavior Data

To support computational detection of cost-presentation bias, the framework integrates the structural stream and behavioral stream into a unified feature representation. As shown in Figure 1, the input layer consists of page snapshots, DOM trees, and timestamped interaction logs. The structural stream extracts 24 interface features from each page, including cost-field position, DOM hierarchy depth, first-screen exposure, prompt completeness, font scale, contrast ratio, button–cost distance, cancellation-entry level, and hidden expansion count. These variables are normalized and projected into a 128-dimensional structural vector. The behavioral stream encodes six types of interaction events, namely click, dwell, backtrack, scroll, expansion, and abandonment search, into a 256-step temporal sequence with a 200 ms time window. After timestamp synchronization, session sampling, missing-step padding, and z-score normalization, the behavioral sequence is mapped into a 64-dimensional behavior vector. The two streams are then integrated through cross-modal concatenation and attention-based alignment using user ID, page ID, and step index as alignment identifiers. The resulting joint input tensor provides the computational basis for decision-bias recognition and cancellation-path complexity analysis [3-4].

3.2 Decision Bias Recognition Model Based on Interface Structural Features

Based on the joint input tensor generated from interface parsing and clickstream alignment, the decision-bias recognition module adopts a parallel encoding structure for the structural branch and the behavioral branch [5-6]. The structural branch receives a 24-dimensional interface vector, including DOM hierarchy depth, cost-field first-screen exposure, price-block coordinates, font scale, contrast ratio, button–cost distance, cancellation-entry depth, hidden-menu expansion count, and prompt completeness. These variables are normalized and projected into a latent structural representation so that cost visibility, spatial prominence, interface hierarchy, and visual-load pressure can be jointly represented. It should be noted that the model does not claim to directly capture gaze fixation or real-time attention allocation. Instead, it estimates decision-bias risk by combining layout-based salience proxies with behavioral hesitation signals, such as prolonged dwell time, repeated scrolling,

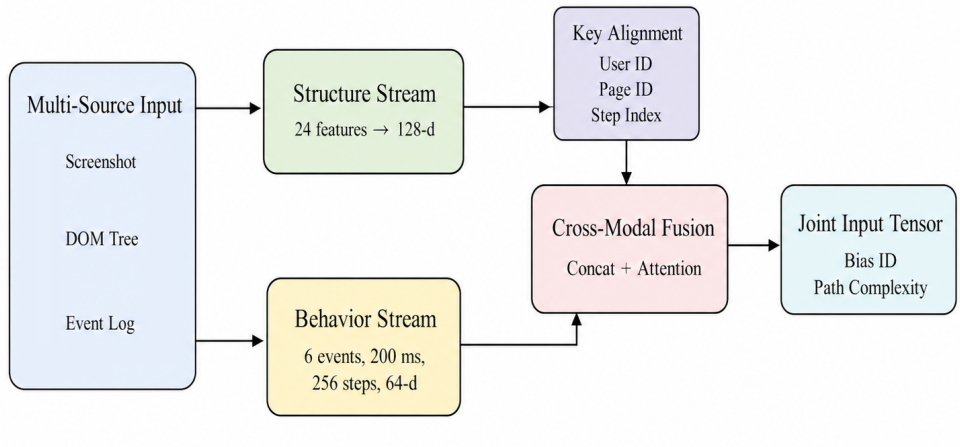


Figure 1: Framework diagram for the integration of cost presentation structure and user behavior data

backtracking, and expansion of hidden fields. This design makes the model suitable for large-scale interface auditing where eye-tracking data are usually unavailable, while still acknowledging that heuristic layout indicators cannot fully replace physiological attention measurements. The behavioral branch receives a 256-step operation sequence with a 200 ms step width, where click, dwell, scroll, back-navigation, field expansion, and abandonment-search events are encoded according to their temporal order. This branch captures user hesitation, repeated navigation, and delayed recognition of cancellation entries.

A context vector is further introduced to calibrate page-specific differences. The context vector consists of cost first-screen exposure rate, prompt expansion frequency, and back-navigation depth, which are directly related to cost discoverability and navigation friction. The structural representation, behavioral representation, and context vector are integrated through vector concatenation, linear projection, and structure–behavior alignment. The bias probability is defined as:

$$p_i = \sigma\left(W_o\left[\alpha_i h_i^s \,\|\, \beta_i h_i^b \,\|\, \gamma_i c_i\right] + b_o\right), \alpha_i soft\max\left(\frac{Q_i^s K_i^{bT}}{\sqrt{96}}\right) \quad (1)$$

where i denotes the i th subscription session sample, p_i denotes the decision bias probability for that session, σ is the Sigmoid activation function, $\|$ denotes vector concatenation, Q_i^s and K_i^b are the linear projections of the structural branch query matrix and behavioral branch key matrix, respectively, and α_i denotes the structure-behavior alignment weight, β_i denotes the temporal gating coefficient, which characterizes the sensitivity to dwell, scroll, and click variations, γ_i denotes the interface context calibration coefficient,$W_o \in R^{1\times 224}$ and b_o denote the output layer weights and bias, respectively; this output is subsequently used as prior input for path complexity modeling.

3.3 Path Complexity Modeling Method for Cancellation Behavior Prediction

After obtaining the session-level decision bias prior p_i , cancellation behavior is no longer counted on a per-step basis but is represented as a directed path graph $G_i = (V_i, E_i)$, where the set of nodes V_i corresponds to interface states such as the account page, subscription page, help page, and secondary confirmation page, and the set of edges E_i represents state transitions triggered by clicks, back navigation, expansion, redirection, and search. In this study, graph-based indicators are not treated as purely mathematical descriptors. They are used as computational proxies for task-completion difficulty. A longer node sequence reflects a greater number of required actions, branch entropy reflects uncertainty in choosing the next operation, backtracking reflects correction after navigation failure, and accumulated dwell time reflects hesitation during the cancellation task. Therefore, path complexity is interpreted as the structural resistance encountered by users when trying to complete a goal-directed cancellation process. Path complexity is defined as:

$$C_i = \lambda_1 \frac{|V_i|}{V_{\max}} + \lambda_2 \frac{\sum_{(u,v)\in E_i} w_{uv}}{E_{\max}} + \lambda_3 H_i + \lambda_4 R_i + \lambda_5 \frac{T_i}{T_{\max}} \quad (2)$$

where C_i is the aggregate complexity of the i th session cancellation, $|V_i|$ is the number of visited nodes, $V_{\max}$ is the maximum sample node size, w_{uv} is the transition cost of edge (u, v), $E_{\max}$ is the upper bound of normalized edge counts, H_i is the path branch entropy, which characterizes the uncertainty of entry point branching, R_i is the back-to-top rate, T_i is the cumulative dwell time, $T_{\max}$ is the time-normalized constant, and $\lambda_1 \sim \lambda_5$ is the complexity weight. Further, let:

$$w_{uv} = \eta_1 \Delta d_{uv} + \eta_2 b_{uv} + \eta_3 x_{uv} + \eta_4 \frac{\tau_{uv}}{\tau_{\max}} + \eta_5 m_{uv} \quad (3)$$

where Δd_{uv} represents the page hierarchy difference, b_{uv} indicates whether a back operation occurred, x_{uv} indicates whether a hidden item was expanded, τ_{uv} is the edge dwell time, $\tau_{\max}$ is the maximum edge time, and $m_{uv}$is the semantic jump penalty term. The semantic jump penalty is used to measure non-direct transitions such as "Subscription Management—Help Center—Customer Service Page." This term is introduced because such transitions interrupt the continuity between the user's cancellation goal and the visible operation path. When a user is redirected from a subscription-management page to a help center or customer-service page, the information scent of the

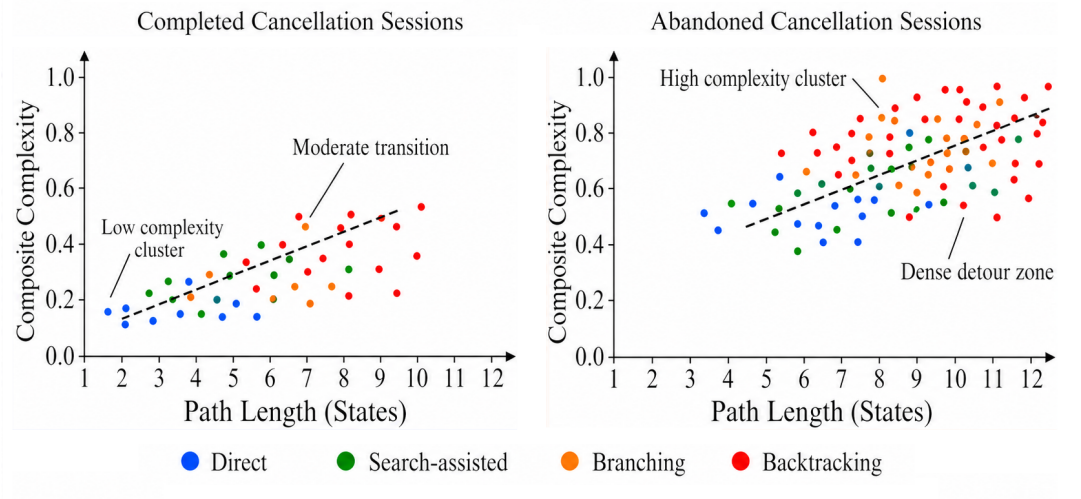


Figure 2: Side-by-side scatter plot of path length and aggregate complexity

original task becomes weaker, and the perceived distance to task completion increases. Thus, the semantic jump penalty represents goal-path discontinuity rather than a simple increase in page count. The probability of cancellation or abandonment is then expressed as:

$$\hat{y}_i = \sigma(\theta_1 C_i + \theta_2 p_i + \theta_3 S_i + \theta_4 Q_i + b) \tag{4}$$

Here, $\hat{y}_i$ represents the predicted probability of cancellation or abandonment, S_i represents the number of retrievals,Q_i represents the number of secondary confirmation triggers, b represents the bias term, and$\theta_1 \sim \theta_4$ represents the regression coefficient; These graph-based variables should be understood as observable proxies of perceived difficulty rather than direct psychological measurements. Their validity is supported in this paper through their association with cancellation abandonment, while future studies may further verify them using subjective workload scales or post-task difficulty ratings. To visually represent the distribution in the complexity space, Figure 2 can be plotted, using a side-by-side scatter plot to illustrate the mapping relationship between session-level path length L_i and comprehensive complexity C_i.

3.4 Modeling Cost-Ahead and Path Compression for Interface Optimization Interventions

Based on the deviation probability p_i and path complexity C_i , the interface optimization intervention module unifies cost-upfront display and cancellation path compression into a solvable interface configuration problem. It imposes common restrictions on the first screen visibility ratio of cost text, pixel distance between price description and confirmation button, cancel entry level, confirm node count, secondary retention drop trigger count, and keyfields. When the first screen visibility region is coded as a continuous value from 0 to 1, the distance between the button and the cost block is discretized in 16 pixel increments, the path hierarchy is restricted to 1-4, and the number of validation nodes is controlled from 2 to 5 [7]. The intervention objective is defined as:

$$\min J_i = \rho_1(1-\varphi_i) + \rho_2 \frac{D_i}{D_{\max}} + \rho_3 \frac{L_i}{L_{\max}} + \rho_4 \frac{R_i}{R_{\max}} + \rho_5 \frac{M_i}{M_{\max}} \tag{5}$$

where J_i represents the intervention cost for the i th subscription interface, φ_i represents the first-screen exposure rate of cost information, D_i represents the visual distance between the cost description and the primary action button, L_i represents the cancellation link step size, R_i represents the number of back button triggers, M_i represents the number of hidden menu or secondary expansion instances, $D_{\max}, L_{\max}, R_{\max}, M_{\max}$ are the normalized upper bounds for each metric, and $\rho_1 \sim \rho_5$ is the weighting coefficient; Based on this, It is possible to jointly rearrange page layout parameters, prompt locations, and path nodes, so that the experimental part can assess the effectiveness of the interface interventions under the same constraints.

4 Experimental Results and Analysis

4.1 Experimental Design

The experimental dataset contains 4,276 subscription-related interface pages and their corresponding user interaction sequences collected from subscription scenarios on Amazon- and Shopify-based platforms. The page set includes subscription entry pages, checkout pages, cancellation pages, and secondary confirmation pages. For each page, the proposed parser extracts DOM hierarchy, visible cost blocks, action-button coordinates, cancellation-entry depth, hidden-panel state, and visual salience indicators. For each interaction session, the behavioral log records click events, dwell duration, scrolling depth, back-navigation count, cancellation-entry search delay, path-jump length, field re-expansion frequency, and final abandonment state [8]. The original clickstream was sampled at 100 ms intervals, and each valid interaction session was defined as one continuous task record from entry into a subscription-related page to payment confirmation, successful cancellation, or cancellation abandonment. The 3,892 valid sessions were generated by real participants rather than simulated agents or automated scripts. All

Table 1: Comparison of Misjudgment Rates for Payment Amounts Under Different Cost Presentation Methods

Cost Presentation Method	Number of Session Samples	Mean Misjudgment Rate (%)	Standard Deviation (%)	95% Confidence Interval
Deep Interface (Original Structure)	1,946	22.4	3.6	[21.2, 23.6]
Pre-rendering + Path Compression (Optimized Structure)	1,946	8.1	2.3	[7.4, 8.8]

participants provided informed consent before the experiment and were told that the study concerned subscription-interface usability; however, the specific hypotheses regarding cost placement and cancellation friction were disclosed only after task completion to reduce potential Hawthorne effects. Sessions with operation intervals longer than 2.5 s, incomplete page identifiers, missing terminal states, or repeated abnormal refresh behavior were removed. Basic demographic and device information, including age group, gender, prior subscription experience, device category, operating system, and browser type, was recorded in anonymized form and used for stratified balance checking. The dataset was then divided by user identity into training, validation, and testing subsets with proportions of 70%, 15%, and 15%, respectively, to avoid cross-session leakage from the same participant.

Non-subscription browsing records are added as auxiliary covariates to reduce the influence of general browsing habits and platform usability differences. All numerical variables are normalized by z-score transformation before model training. Five-fold cross-validation is used to evaluate the stability of the decision-bias detection model and the cancellation-abandonment prediction model. Because the dataset was based on observed interaction records rather than a randomized A/B experiment, the interface comparison was treated as a matched observational analysis. To reduce confounding effects, sessions were matched or stratified according to prior subscription experience, platform type, device category, browser type, session duration, entry page type, and baseline browsing intensity. Non-subscription browsing records were also used as auxiliary covariates to control general navigation habits and platform usability differences. During interface comparison, the cost first-screen exposure ratio, button–cost block distance, cancellation-path depth, and confirmation-node number were evaluated under the same constraint settings, but the results were interpreted as controlled associations rather than strict causal effects. The optimizer was Adam, the initial learning rate was 0.001, and the batch size was fixed at 128.

4.2 Analysis of Payment Amount Misclassification Rates Under Different Cost Presentation Methods

On the basis of the 3,892 effective sessions determined in the experimental design, the samples were classified into a deep interface group (cost information not visible on the first screen, exposure rate < 0.3) and a front display group (exposure ratio ≥ 0.8), and at the same time, the interference conditions of path compression (cancellation depth ≤ 4 levels) and visual reinforcement (button – cost block distance ≤ 48 px) [9]. Table 1 presents the statistical results of users' misjudgment rates regarding the final payment amount under different cost presentation methods.

Table 1 shows that both groups contain 1,946 valid session samples, which ensures a balanced comparison and reduces sample-size bias. Under the deep-interface original structure, the mean misjudgment rate of the final payment amount reaches 22.4%. From a computational perspective, this group is characterized by lower first-screen exposure, larger DOM hierarchy depth, longer button–cost distance, and more hidden expansion states, which jointly increase the structural feature values associated with cost invisibility. After applying cost front-loading, path compression, and visual salience enhancement, the mean misjudgment rate decreases to 8.1%. The absolute reduction is 14.3 percentage points, and the relative decrease is approximately 63.84%. This result suggests that, after controlling for prior subscription experience, device type, platform source, and baseline browsing behavior, lower DOM depth, higher first-screen exposure, shorter button–cost distance, and fewer hidden expansion states are consistently associated with a lower payment misclassification rate. Therefore, the observed reduction should be interpreted as a robust structural association rather than conclusive causal evidence from randomized interface assignment.

The standard deviation also reflects a clear improvement. The deep-interface group reports a standard deviation of 3.6%, whereas the optimized structure group records a lower value of 2.3%, representing a decrease of 1.3 percentage points. This means that the optimized presentation method not only reduces the overall misclassification level but also improves the consistency of user judgment across sessions. The 95% confidence intervals further confirm this distinction. The confidence interval for the deep-interface group is [21.2, 23.6], while that of the optimized structure group is [7.4, 8.8]. Since the two intervals do not overlap, the difference between the two presentation methods shows strong statistical separation. Overall, hiding cost information in deeper interface layers significantly increases the risk of payment amount misjudgment, whereas front display, cancellation path compression, and keeping the button–cost block distance within 48 px effectively reduce users' cognitive bias and improve payment decision accuracy.

4.3 Analysis of the Impact of Cancellation Process Complexity on Abandonment of Cancellation Attempts

On the basis of the error rate analysis, we further investigate the influence of cancellation process complexity on cancellation abandonment behavior [10]. Using the path complexity computation formula, the 1,946 cancellation sessions were split into a low complexity group (complexity < 0.42, equivalent to a cancellation path

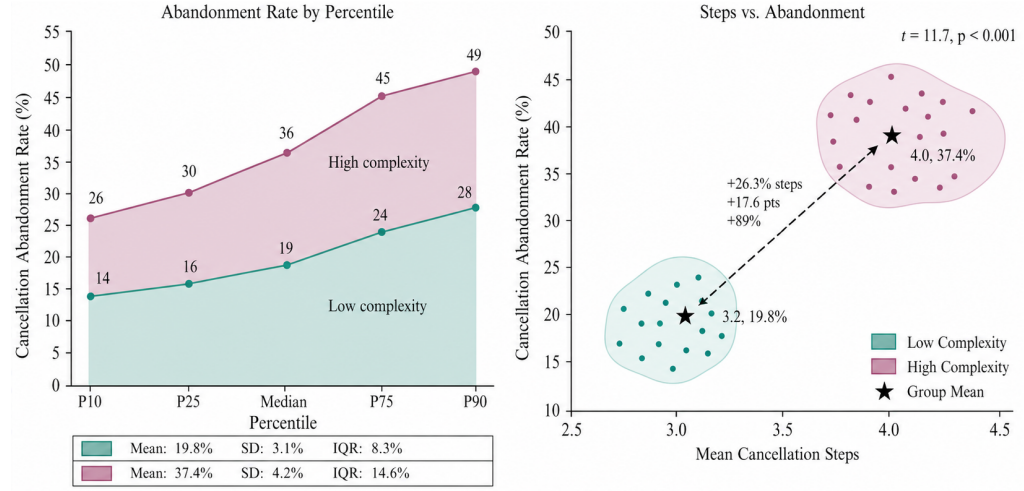


Figure 3: Comparison of Abandonment Rates Under Different Cancellation Process Complexities

length ≤ 4 steps) and a high complexity group (complexity ≥ 0.42, equivalent to a cancellation path length that is approximately 25% longer than the original baseline). Figure 3 shows the combined results of the cancellation abandonment rates for both groups.

As shown in Figure 3, the average rate of abandonment was 37.4% (SD 4.2%, I-QUARTILE 14.6%), and that of low complexity was 19.8% (SD 3.1%, interquartile 8.3%). There was a significant difference ($t = 11.7$, $p < 0.001$), and the rate of abandonment was 17.6 percent higher in the high-complexity group than in the low-complexity group, which was about 89% higher. In addition, the average number of cancellation steps in this group increased by 26.3% (from 3.2 to 4.0) compared with the low complexity group, with an increase of approximately 25% in the cancellation step and a significant increase in the abandonment rate.

4.4 Comparison of User Decision-Making Behavior Before and After Interface Structure Optimization

After discovering the influence of cancellation complexity on abandonment behavior, we compared the performance of the user decision before and after the optimization of the interface structure. Based on the intervention goal, a combined intervention consisting of upfront cost presentation (first screen exposure ratio ≥ 0.8), path compression (cancellation step ≤ 4 levels), and visual enhancement (distance from button to cost block ≤ 48 px) was applied to 1,946 sessions in the trial set, and another 1,946 sessions were used as a control group. Figure 4 shows a combined comparison of the precision of the decision and the unscheduled subscription rate before and after optimization.

Figure 4 further shows that the optimized interface group is associated with better payment-related decision performance than the original interface group under matched observational conditions. Although the comparison controls several observable confounders, such as prior subscription familiarity, device category, platform source, and session-level browsing intensity, unobserved factors, including temporary distraction or individual price sensitivity, may still affect the results. Therefore, the findings support the practical relevance of cost front-loading, path simplification, and visual distance control, but they should not be overinterpreted as fully randomized A/B-test evidence. Since both the optimized group and the original interface group contain 1,946 sessions, the comparison is not affected by an imbalance in sample size. In terms of decision accuracy, the optimized group reaches an average value of 91.9%, whereas the original group records only 77.6%. The absolute difference is 14.3 percentage points, and the relative improvement is approximately 18.43% when the original group is used as the baseline. This result suggests that when cost information is presented on the first screen, the cancellation path is compressed to no more than four levels, and the distance between the button and the cost block is controlled within 48 px, users can identify the actual payment obligation earlier and make more accurate subscription decisions.

The dispersion indicators also show that the optimized interface improves decision stability. The standard deviation of decision accuracy decreases from 3.5% in the original group to 2.7% in the optimized group, representing a reduction of 0.8 percentage points, or about 22.86%. The interquartile range also narrows from 6.1% to 4.6%, indicating that the optimized structure not only increases the average accuracy level but also reduces variation across users. The change in the unplanned subscription rate provides a more direct behavioral outcome. The original group shows an unplanned subscription rate of 12.7%, while the optimized group decreases to 6.3%, with an absolute reduction of 6.4 percentage points and a relative decrease of about 50.39%. In addition, the standard deviation falls from 2.8% to 1.9%, and the interquartile range decreases from 4.2% to 2.8%. These results indicate that the optimized interface reduces the risk of accidental subscription caused by hidden cost information and overly complex cancellation paths. Overall, the optimization does not merely increase information visibility; it reduces cognitive load through cost front-loading, path simplification,

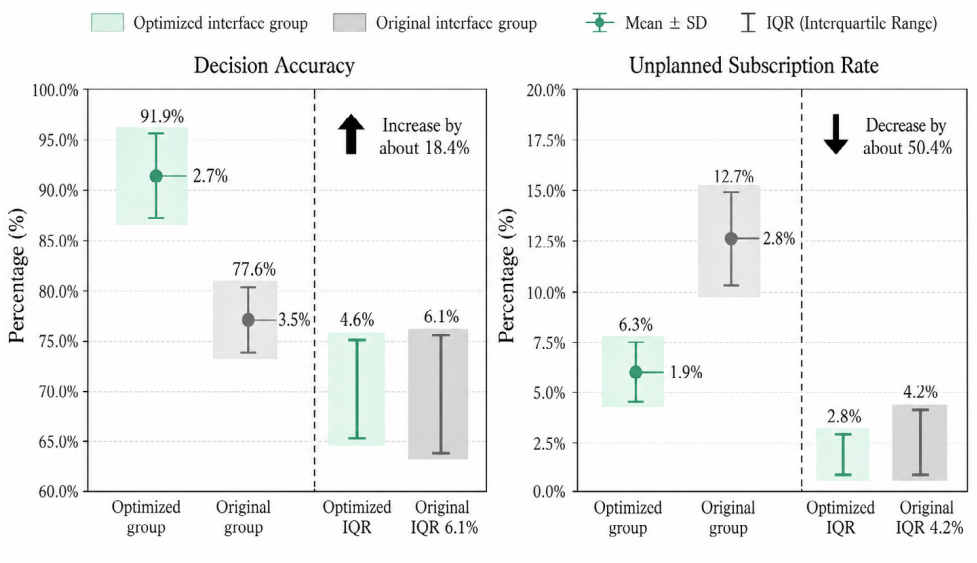


Figure 4: Combined Comparison of User Decision-Making Performance Before and After Interface Structure Optimization

and visual distance control, thereby improving both the accuracy and consistency of user decision-making.

5 Conclusions

This study proposes a computational framework for detecting decision bias and cancellation friction caused by cost presentation in online subscription interfaces. By transforming subscription pages into structured interface objects, the framework extracts DOM hierarchy, first-screen cost exposure, visual salience, button–cost distance, cancellation-entry depth, hidden expansion states, and clickstream behavior sequences. A structure–behavior fusion model is used to estimate session-level decision-bias probability, while a directed cancellation-path graph quantifies process complexity through node transitions, edge costs, branch entropy, dwell-time accumulation, backtracking, and semantic-jump penalties. The experimental results show that deep cost placement increases payment misclassification, whereas cost front-loading, visual salience enhancement, and path compression reduce the mean misjudgment rate from 22.4% to 8.1%. High-complexity cancellation paths produce a 37.4% abandonment rate, while low-complexity paths reduce the rate to 19.8%. After interface optimization, decision accuracy increases from 77.6% to 91.9%, and unplanned subscription decreases from 12.7% to 6.3%. These findings indicate that subscription-interface fairness can be evaluated through measurable computational features rather than only through post-hoc user surveys or qualitative descriptions. Although demographic and device-level metadata were used for stratified balance checking, the dataset was still limited to controlled subscription tasks on Amazon- and Shopify-based interfaces; therefore, future work should further validate the framework across broader age groups, mobile-only users, and different subscription service categories.

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